exp0526: Pure 0522-Style Closed-Loop Controller Learning for Age-Structured Predator–Prey PDEs

exp0526 notes

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1 Purpose

This note reformulates the age-structured predator–prey control setting from *Lotka–Sharpe Neural Operators for Control of Population PDEs* as a pure exp0522-style closed-loop controller learning problem [1].

The first exp0526 version should not learn a Lotka–Sharpe neural operator and should not approximate the paper's analytic controller. Instead, the model directly outputs a scalar control signal $u(t)$, applies it to the population dynamics, and receives loss from the resulting distance to the equilibrium state.

The controller interface is

$$\big(\eta_1(t), \eta_2(t), 1;\ \text{internal self-talk}\big) \longmapsto u(t).$$

The ordinary external observation is (η_1, η_2) together with a constant pulse 1. The self-talk state is internal to the exp0522 model: the previous language output is fed into the next step.

2 Problem From the Miroslav Paper

The environment is the two-species age-structured predator–prey PDE used in the Lotka–Sharpe neural-operator paper and inherited from the age-structured predator–prey stabilization literature [1, 4]. It is a McKendrick–von Foerster type age-structured model [2]:

$$\partial_t x_1(a,t) + \partial_a x_1(a,t) = -x_1(a,t)\left[\mu_1(a,t) + u(t) + \int_0^A g_1(\alpha) x_2(\alpha,t)\,\mathrm{d}\alpha\right], \tag{1a}$$

$$\partial_t x_2(a,t) + \partial_a x_2(a,t) = -x_2(a,t)\left[\mu_2(a,t) + u(t) + \frac{1}{\int_0^A g_2(\alpha) x_1(\alpha,t)\,\mathrm{d}\alpha}\right], \tag{1b}$$

$$x_i(0,t) = \int_0^A k_i(a,t) x_i(a,t)\,\mathrm{d}a, \qquad i = 1,2. \tag{1c}$$

Here $x_i(a,t) > 0$ are the age profiles, k_i are fertility profiles, μ_i are mortality profiles, g_i are interaction kernels, and $u(t) \geq 0$ is the scalar dilution or harvesting input.

The control target is a positive equilibrium profile

$$x_i(a,t) \to x_i^*(a,t), \qquad i = 1,2,$$

or, equivalently for the reduced controller coordinates, the equilibrium

$$\eta_1(t) \to 0, \qquad \eta_2(t) \to 0.$$

3 Equilibrium Error Coordinates

For each species, the Lotka–Sharpe scalar is defined implicitly by the classical Lotka–Sharpe age-distribution condition [3]:

$$\int_0^A k_i(a,t)\exp\left(-\int_0^a \big(\mu_i(s,t)+\zeta_i(t)\big)\,\mathrm{d}s\right)\mathrm{d}a = 1. \tag{2}$$

The Lotka–Sharpe neural-operator paper uses this quantity to construct transformed coordinates [1]

$$\eta(t) = (\eta_1(t), \eta_2(t)).$$

In the paper's notation,

$$\eta_i(t) = \log\left(\frac{\langle \pi_{0,i}(\cdot,t), x_i(\cdot,t)\rangle}{x_i^*(0,t)\kappa_i(t)}\right). \tag{3}$$

The exp0526 controller observes η_1, η_2, not the full hidden ecological parameter state. The full PDE remains the environment, while η is the low-dimensional control error exposed to the agent.

4 What We Change

The Lotka–Sharpe neural-operator paper keeps the analytic feedback law and uses learning only to approximate the implicit Lotka–Sharpe map

$$(k,\mu) \mapsto \zeta.$$

That is not the first exp0526 task.

By contrast, the first exp0526 task is direct closed-loop controller learning:

$$\big(\eta_1(t), \eta_2(t), 1;\ m_{t-1}\big) \longmapsto u(t),$$

where m_{t-1} is the model's own previous language output. The model is judged by what its control does to the environment, not by whether it matches an analytic control formula.

Problem 1 (Pure exp0526 closed-loop controller)**.** *Train an exp0522-style self-talk MLP inside the full two-species age-structured PDE environment* (1)*. At each time step the agent observes $\eta_1(t)$, $\eta_2(t)$, and a constant pulse* 1*, produces a scalar control $u(t)$, and is optimized through the resulting future equilibrium error.*

5 0522 Mechanism Reused

The exp0522 mechanism is not an external memory module. It is a recurrent self-talk loop created by feeding the previous language output into the next MLP step.

At step t, the hidden network computes

$$h_t = F_\theta\big(\eta_1(t), \eta_2(t), 1, e_{t-1}, m_{t-1}\big),$$

then emits both a control preactivation and a new language state:

$$\tilde{u}_t = W_u h_t + b_u, \tag{4}$$

$$m_t = Rh_t. \tag{5}$$

Here R is the fixed sparse signed readout used by exp0522. The next step receives m_t. This is the sense in which the model has memory: the language output from the previous step becomes part of the next step's input.

The ordinary problem statement remains

$$(\eta_1, \eta_2, 1) \mapsto u,$$

because m_t is an internal model state rather than an externally measured ecological variable.

Error Channel

In exp0522 prediction, the error channel is a scalar prediction error. In exp0526 control, the first version should reinterpret the error channel as a control-error view:

$$e_t = \eta_t = (\eta_1(t), \eta_2(t)).$$

This is not a control-label error. It is the current distance direction from equilibrium, made available through the same error-feedback interface so that the 0522 ablations remain meaningful.

The no-error comparison forces this channel to zero. The no-language comparison uses $m_t \equiv 0$. Blink and stutter tests temporarily remove the error or language pathway during rollout.

6 Closed-Loop Learning Objective

The network produces an unconstrained scalar $\tilde{u}_t$. The actual control applied to the environment is

$$u_t = \text{softplus}(\tilde{u}_t),$$

which enforces $u_t \geq 0$ while preserving smooth gradients.

For one differentiable rollout window of length T, the main objective is distance to equilibrium:

$$\mathcal{L}_\eta = \frac{1}{T} \sum_{t=1}^{T} \left(\eta_1(t)^2 + \eta_2(t)^2 \right). \tag{6}$$

A small control-energy regularizer can be added:

$$\mathcal{L} = \mathcal{L}_\eta + \lambda_u \frac{1}{T} \sum_{t=1}^{T} u_t^2, \qquad 0 < \lambda_u \ll 1. \tag{7}$$

The required gradient path is

$$\mathcal{L} \to \eta_{t+1} \to x(\cdot, t+1) \to u_t \to \tilde{u}_t \to h_t \to \theta.$$

Because $m_t = Rh_t$ is fed to later steps, gradients inside a training window also flow through the self-talk loop. At window boundaries, the message and PDE state are carried forward but detached, matching the continuous-window protocol from exp0522.

7 Differentiable PDE Tooling

The first implementation should use a PyTorch differentiable age-grid solver. Let

$$a_j = j\Delta a, \qquad j = 0, \dots, N_a.$$

Use a first-order upwind update for the transport term:

$$\frac{x_{i,j}^{n+1} - x_{i,j}^{n}}{\Delta t} + \frac{x_{i,j}^{n} - x_{i,j-1}^{n}}{\Delta a} = -x_{i,j}^{n}\,\mathcal{R}_{i,j}^{n}, \qquad j \geq 1, \tag{8}$$

where $\mathcal{R}_{i,j}^{n}$ contains mortality, control, and interaction terms. The boundary is updated by quadrature:

$$x_{i,0}^{n+1} = \sum_{j=0}^{N_a} k_i(a_j, t_n) x_{i,j}^{n} \Delta a. \tag{9}$$

All state updates, quadratures, η computation, softplus control, and losses should remain in PyTorch tensors so that the equilibrium loss can backpropagate through the rollout to the controller.

8 Time-Varying Ecology

The ecology should not be static. The first version uses the same profile families used in the Lotka–Sharpe neural-operator experiments [1]:

$$k(a,t) = k_{\text{base}} + k_{\text{amp}}(t) \exp\left(-\frac{(a - k_{\text{center}})^2}{2k_\sigma^2}\right), \tag{10}$$

$$\mu(a,t) = \mu_{\text{base}}(t) + \mu_{\text{juv,amp}} e^{-\mu_{\text{juv}} a} + \mu_{\text{sen,amp}}(t) a^{\mu_{\text{sen}}}, \tag{11}$$

$$g(a) = g_{\text{base}} + g_{\text{amp}} \exp\left(-\frac{(a - g_{\text{center}})^2}{2g_\sigma^2}\right). \tag{12}$$

The centers, widths, and interaction profile are fixed in the first version. The ecological regime changes through mixed-sin modulation of amplitude-like parameters:

$$k_{\text{amp}}(t), \qquad \mu_{\text{base}}(t), \qquad \mu_{\text{sen,amp}}(t).$$

A generic mixed-sin modulation is

$$q(t) = q_0 \left[1 + \rho \frac{\sum_{\ell=1}^{L} A_\ell \sin(\omega_\ell t + \varphi_\ell)}{\sum_{\ell=1}^{L} |A_\ell|}\right], \qquad 0 < \rho < 1. \tag{13}$$

The model does not observe $q(t)$, $k(a,t)$, $\mu(a,t)$, or $\zeta(t)$. It only observes $\eta_1(t)$, $\eta_2(t)$, and the constant pulse. The language loop is therefore tested as an internal state that may help track hidden ecological regime changes.

9 Continuous Training and Testing

Training should follow the exp0522 continuous case:

- time moves forward in one long ecological stream;

- training windows are consecutive pieces of this stream;
- the PDE state, language state, and error view are carried forward;
- the carried states are detached at window boundaries;
- the stream is not rewound to $t = 0$ after each update.

Evaluation should reserve a future segment of the same ecological stream. The model trains on the earlier segment, then rolls into the held-out future segment without weight updates. This tests whether the learned closed-loop controller continues to stabilize the ecology as the hidden birth and mortality parameters keep evolving.

10 First Experiment Design

The first experiment should compare:

1. **full self-talk controller:** η, pulse, error view, and language all live;
2. **no-language controller:** same external observations, but $m_t \equiv 0$;
3. **no-error-view controller:** language live, but $e_t \equiv 0$;
4. **minimal controller:** no language and no error view, only $(\eta_1, \eta_2, 1)$;
5. **blink/stutter rollouts:** temporary removal of error or language during evaluation.

The primary metrics are:

- mean and final $\eta_1^2 + \eta_2^2$;
- mean control energy u^2;
- positivity of $x_i(a, t)$ and $u(t)$;
- future-segment stability after training stops;
- recovery after blink/stutter interruptions.

11 Working Summary

The paper supplies a biologically meaningful full PDE environment and equilibrium coordinates. Exp0522 supplies the self-talk MLP and continuous rollout protocol. Exp0526 combines them by training a controller directly from closed-loop equilibrium loss:

$$\big(\eta_1(t), \eta_2(t), 1; \text{self-talk}\big) \mapsto \tilde{u}_t \mapsto u_t = \text{softplus}(\tilde{u}_t) \mapsto x(\cdot, t+1) \mapsto \eta(t+1) \mapsto \mathcal{L}_\eta.$$

References

[1] Miroslav Krstić, Iasson Karafyllis, Luke Bhan, and Carina Veil. Lotka–sharpe neural operators for control of population PDEs, 2026.

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[3] F. R. Sharpe and A. J. Lotka. A problem in age-distribution. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 21(124):435–438, 1911.

[4] Carina Veil, Miroslav Krstić, Iasson Karafyllis, Mamadou Diagne, and Oliver Sawodny. Stabilization of predator–prey age-structured hyperbolic PDE when harvesting both species is inevitable. *IEEE Transactions on Automatic Control*, 71(1):123–138, 2026.